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Key to the multiple choice questions

1. $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

Content	CH_4	O_2	CO_2
Stoichiometric coefficient	1	2	1
Volume of reactants allowed to react	40 mL	80 mL	
Volume of reactant reacted and product formed	40 mL	80 mL	40 mL
Volume of gas after cooling to the room temperature	-	-	-

Since the product was cooled to room temperature, water exists mostly as liquid. Hence, option (a) is correct

2.

$$\text{200 u} = \frac{(200\times 90)+(199\times 8)+(202\times 2)}{100} = 199.96$$

3. Correct reason: Total number of entities present in one mole of any substance is equal to 6.022×10^{23}.

4. Reaction 1 : $2\,C + O_2 \rightarrow 2\,CO$

2×12 g carbon combines with 32 g of oxygen. Hence,

Equivalent mass of carbon $= \frac{2\times 12}{32}\times 8 = 6$

Reaction 2 : $C + O_2 \rightarrow CO_2$

12 g carbon combines with 32 g of oxygen. Hence,

Equivalent mass of carbon $= \frac{12}{32}\times 8 = 3$

5. Let the trivalent metal be M^{3+}

Equivalent mass = mass of the metal / valance factor

$9\ g\ eq^{-1}$ = mass of the metal / 3 eq

Mass of the metal = 27 g

Oxide formed M_2O_3 ;

Mass of the oxide = (2 x 27) + (3 x 16)

= **102 g**